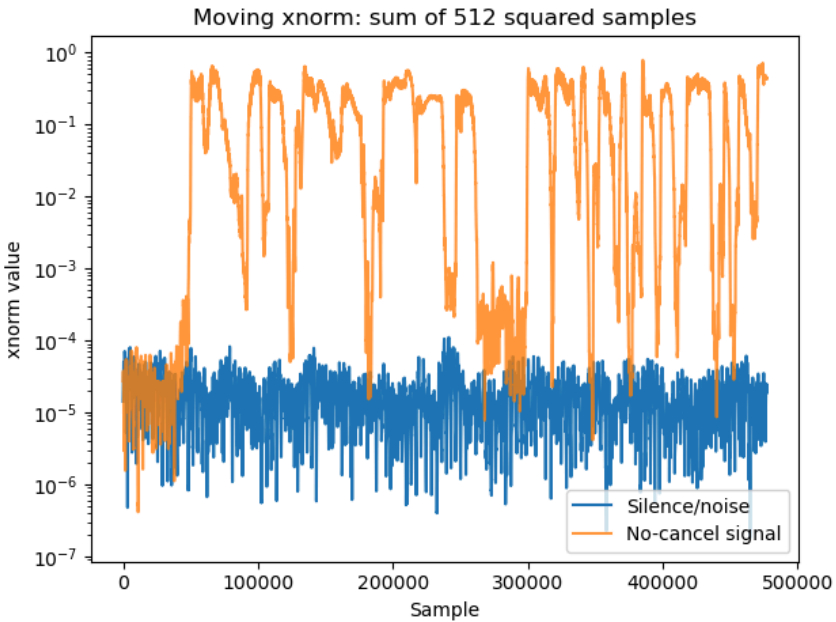
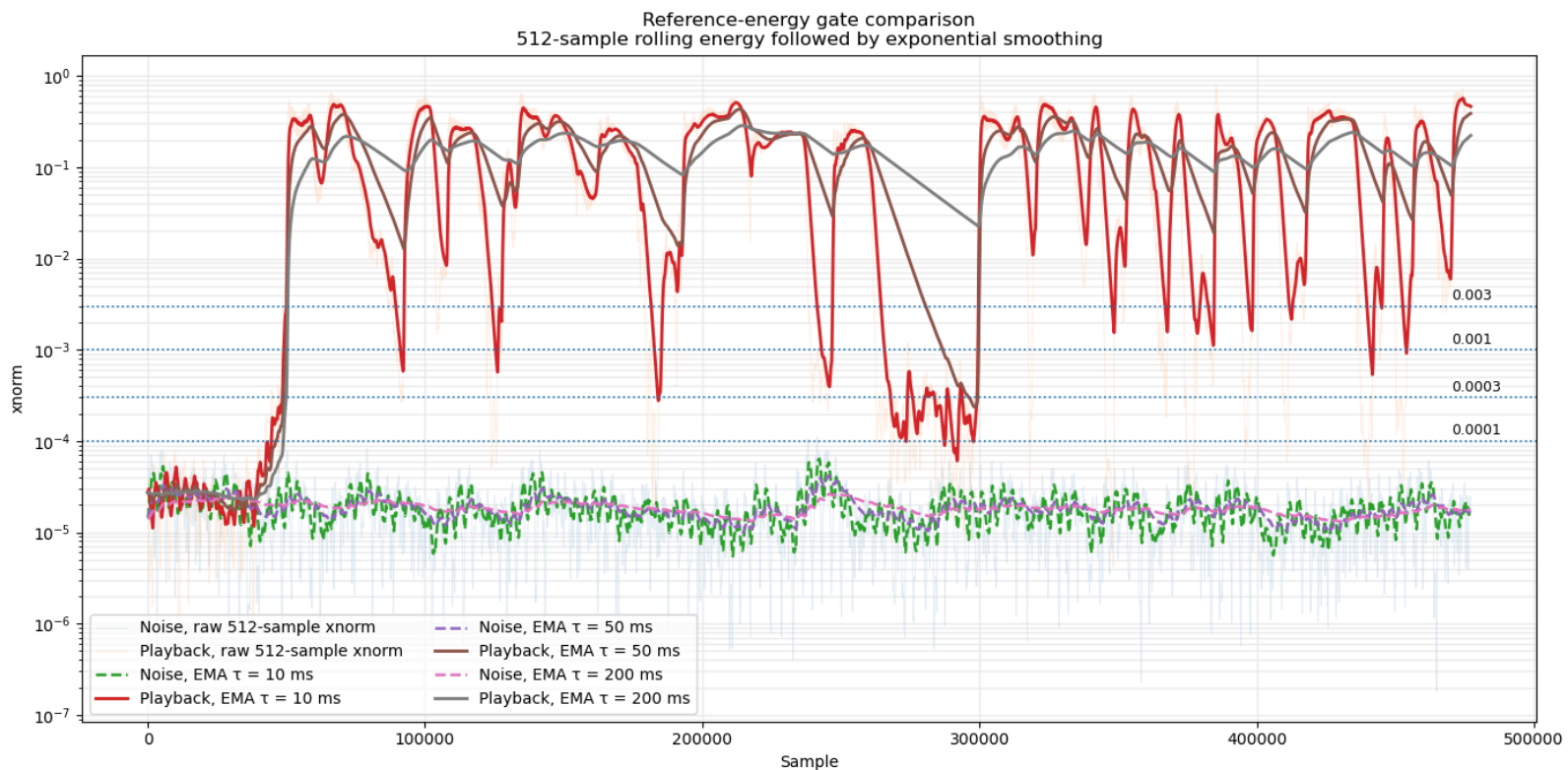


Volume threshold for adapting

- Filter kept adapting to electric and ambient noise
- Coefficients drifted and caused error over time
- No clear cutoff using just the power of the current buffer



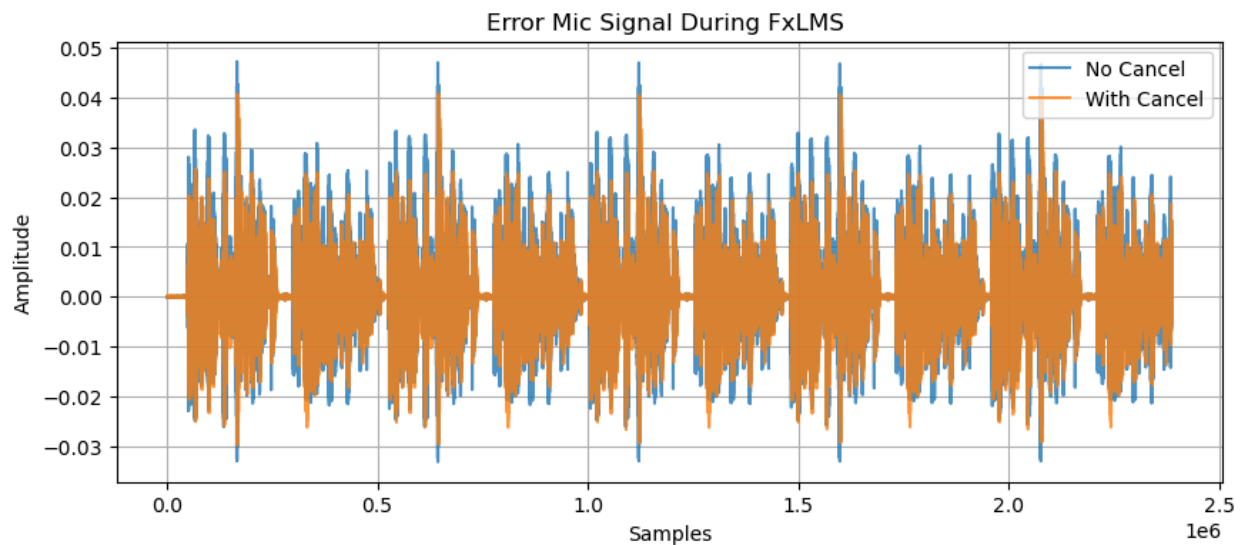
- Moving average did better
 - $y[n] = \alpha x[n] + (1 - \alpha)y[n - 1]$



- Chose characteristic time of 100ms, $\alpha = 0.999896$
 - When input becomes 0, after τ time it degrades to ~36% of its value

Early cancellation results

- 6 iterations, -1.82dB reduction



- Even when it spikes up, it can recover

